GraphHop: Knowledge Graph-Augmented Retrieval for Multi-Hop Question Answering

Lena Okafor

Northbridge University

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Abstract

GraphHop augments Multi-hop QA with a Knowledge Graph built from text. RelForge extracts entity relations, after which GraphHop follows links between question entities to collect a connected evidence set. That set is passed to a RetGen-style generator, allowing the answer to be composed from a path rather than from independently ranked passages. On HotpotQA, GraphHop improves both Exact Match and F1 over RetGen, with the strongest gains on questions requiring three or more hops. The method is motivated by the observation that multi-hop questions are not merely longer retrieval queries: they require identifying intermediate entities and preserving their relationships. GraphHop makes those relationships explicit and uses them to constrain evidence collection.

Keywords

Multi-hop QA, Knowledge Graphs, Evidence paths, HotpotQA

1. Introduction

Multi-hop QA is difficult because a correct answer may depend on a chain of facts distributed across passages. RetGen improves direct retrieval-based answering, but it can miss an intermediate entity or supply disconnected evidence. GraphHop addresses this gap with a Knowledge Graph and a path-aware evidence selector. RelForge provides the graph construction component cited in [1], while RetGen supplies the generator baseline cited in [2].

2. Method

We begin with a corpus-level Knowledge Graph whose nodes are canonical entities and whose edges are relation-bearing triples extracted by RelForge. For each HotpotQA question, entity mentions seed a bounded search. Candidate paths are scored by a combination of semantic question similarity, edge confidence, and path length. We retain a small connected subgraph and map each selected edge back to its supporting passage. The evidence passages are ordered by path position and passed to a RetGen-style generator. A path budget prevents irrelevant expansion, while an answerability gate rejects paths that do not connect the required entities. Training uses answer-bearing paths as positives and disconnected paths as hard negatives. GraphHop therefore uses structure for retrieval while keeping generation text-based.

3. Experiments

On HotpotQA, RetGen reaches 31.2 EM and 42.6 F1. GraphHop reaches 39.4 EM and 50.7 F1, gains of 8.2 and 8.1 points. For questions requiring three or more hops, the EM gain rises to 11.6 points. Most residual errors come from relation extraction misses that break an otherwise valid path.

System	HotpotQA EM	HotpotQA F1
RetGen	31.2	42.6
GraphHop	39.4	50.7

4. Conclusion

GraphHop shows that a Knowledge Graph can turn multi-hop retrieval into explicit path selection. Combining RelForge with a RetGen-style generator yields an eight-point HotpotQA EM improvement and larger gains on long chains. The remaining bottleneck is reliable extraction of every intermediate relation.

References

[1] Raman, P. (2020). RelForge: Neural Relation Extraction for Automatic Knowledge Graph Construction. Conference on Natural Language Systems (CNLS).

[2] Vik, A. (2021). Retrieve-then-Generate: Augmenting Language Models with Retrieved Passages for Knowledge-Intensive Tasks. Conference on Natural Language Systems (CNLS).